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1. (a) Find all the left cosets of the set $\{1, 11\}$ in $U(30)$.

$$\Rightarrow H = \{1, 11\}, G = U(30)$$

$$* U(30) = \{1, 7, 11, 13, 17, 19, 23, 29\}$$

a	$aH \text{ mod } 30$
1	$1\{1, 11\} = \{1, 11\}$
7	$7\{1, 11\} = \{7, 17\}$
11	$11\{1, 11\} = \{11, 1\}$
13	$13\{1, 11\} = \{13, 23\}$
17	$17\{1, 11\} = \{17, 7\}$
19	$19\{1, 11\} = \{19, 29\}$
23	$23\{1, 11\} = \{23, 13\}$
29	$29\{1, 11\} = \{29, 19\}$

- (b) Let n be a positive integer, and let $H = \{0, \pm n, \pm 2n, \pm 3n, \dots\}$. Find all the left cosets of H in $\mathbb{Z}$. How many of these left cosets are there?

$$* G = \mathbb{Z}$$

$$0 + H = H$$

$$1 + H = \{\dots, -n + 1, 1, 1 + n, \dots\}$$

$$2 + H = \{\dots, 2 - n, 2, 2 + n, 2 + 2n, \dots\}$$

$$\vdots$$

$$\Rightarrow n + H = \{\dots, n - 2n, n - n, n, n + n, n + 2n, \dots\}$$

$$= \{\dots, -n, 0, n, 2n, 3n, \dots\} = H$$

By Lemma 3, since $n \in 0 + H$, $n + H = 0 + H = H$. So, after every n cycles, a left coset duplicates H . So, it must be that the number of distinct left cosets are

$$\underbrace{0 + H, 1 + H, 2 + H, \dots, n - 1 + H}_{n \text{ left cosets}}$$

. So, there are $\boxed{n \text{ left cosets of } H \text{ in } \mathbb{Z}.}$

2. Let a and b be non-identity elements of different orders in a group G that has order 155. Prove that the only subgroup of G that contains both a and b is G itself.

Proof. Let G be a group and $a, b \in G$ s.t

$$a \neq e \neq b \quad \text{and} \quad |G| = 155.$$

Let $H \leq G$. We want to show that if $a, b \in H$ then

$$H = G.$$

By Lagrange's Theorem, we know $\frac{|G|}{|H|}$. Since $|G| = 155$, its divisors, which are the possible orders for $|H|$, are 1, 5, 31, and 155. But, by Corollary 2, $|a|$ and $|b|$ must divide both $|H|$ and $|G|$. Since $a, b \in H$, $|H| \neq 1$. We know $|a| \neq |b|$. So, if either $|a| = 155$ or $|b| = 155$, by Corollary 2 that would make $|H|$ divisible by 155. But, if $|H|$ divides $|G|$ and $|G|$ divides $|H|$, by the definition of divisibility

$$|H| = |G|m \text{ and } |G| = |H|n \text{ for some } n, m \in \mathbb{Z}.$$

$$\begin{aligned} \Rightarrow |H| &= (|H|n)m \\ &= |H|(nm) \Rightarrow n = m = 1. \end{aligned}$$

So, this forces $|G| = |H|$, showing that $G = H$. Suppose that

$$|a| = 5 \text{ and } |b| = 31$$

or

$$|a| = 31 \text{ and } |b| = 5.$$

By Corollary 2, $|H|$ must be 155 to divide a and b . Hence, $H = G$. $\square$

3. Let G be a group with $|G| = pq$ where p and q are prime. Prove that every proper subgroup of G is cyclic.

Proof. Let G be a group in which $|G| = pq$ where p and q are prime. Let H be a proper subgroup of G . By Lagrange's Theorem, we know $\frac{|G|}{|H|}$. So, the only divisors of $|G|$, which are the possible orders of H , are 1, p , q and pq . But, since H is a proper subgroup of G ,

$$H \neq G \Rightarrow |H| \neq |G|.$$

So, the only possible orders of H are 1, p and q . The proper subgroup of order 1 is cyclic because $\langle e \rangle = \{e\}$. So, the remaining options are p and q . By Corollary 3, any group of order p or q are also cyclic. $\therefore$ every subgroup of G is cyclic. $\square$

4. Suppose G is a non-Abelian group with $|G| = 8$. Prove that G must have an element of order 4.

Proof. Let G be a non-Abelian group with $|G| = 8$ and $a \in G$. Since G is finite, by Corollary 2, the order of all elements in G must divide $|G|$. The divisors of $|G|$ are 1, 2, 4 and 8. $|a| \neq 8$ because that would imply G is cyclic, which would make it Abelian. Also, $|a| = 1$ implies that a is the identity. Consider a to be any non-identity element of order 2, that is

$$a^2 = e.$$

To prove that G must have an element of order 4, we need to show that elements of order 2 make G Abelian. By HW 3 # 4, groups of order 2 elements are Abelian, which contradicts G being non-Abelian here. So, the only remaining option for $|a|$ is 4. Hence, G must have an element of order 4. $\square$

5. For any integer n greater than 1, let $\phi(n)$ denote the number of positive integers less than n that are relatively prime to n . Prove that if a is any integer relatively prime with n , then $a^{\phi(n)} \bmod n = 1$.

Proof. Let $n \in \mathbb{Z}$ such that $n > 1$ and let $\phi(n)$ denote the number of positive integers less than n that are relatively prime to n . Choose any $a \in \mathbb{Z}$ such that a is relatively prime to n . We want to show that $a^{\phi(n)} \bmod n = 1$. Recall the group $U(n)$ which contains all integers less than and relatively prime to n . We can rewrite the expression $a^{\phi(n)} \bmod n$ as

$$a^{\phi(n)} \bmod n = a^{|U(n)|} \bmod n.$$

But, since a is relatively prime to n , $a \in U(n)$. Since $U(n)$ is finite, by Corollary 4

$$a^{|U(n)|} = e.$$

So,

$$\begin{aligned} a^{\phi(n)} \bmod n &= a^{|U(n)|} \bmod n \\ &= e \bmod n. \end{aligned}$$

But, the identity in $U(n)$ is 1. So,

$$\begin{aligned} a^{\phi(n)} \bmod n &= 1 \bmod n \\ &= 1. \end{aligned}$$

Thus, $a^{\phi(n)} \bmod n = 1$. □

6. Prove that if G is a finite group, the index of $Z(G)$ cannot be prime.

Proof. Let G be a finite group. By contradiction, suppose $|G : Z(G)| = p$ where p is prime. Let $a \in G$ such that $a \notin Z(G)$. Consider $C(a)$, the centralizer of a . We know all elements in $C(a)$ commute with a . Let $b \in Z(G)$ such that b commutes with every element in G . That also means b commutes with a , implying $b \in C(a)$. So, that means

$$Z(G) \leq C(a).$$

By Lagrange's Theorem, $|Z(G)|$ divides $|C(a)|$. Using the definition of divisibility,

$$|C(a)| = |Z(G)|n \text{ where } n \in \mathbb{Z}.$$

Since the centralizer of a group G is a subgroup of G , by Corollary 1,

$$\begin{aligned} |G : C(a)| &= \frac{|G|}{|C(a)|} \\ &= \frac{|G|}{|Z(G)|n} \\ &= \frac{|G|}{|Z(G)|} \cdot \frac{1}{n} \\ &= \frac{p}{n}. \end{aligned}$$

So, $n \mid p$. But since p is prime, either $n = p$ or $n = 1$.

Case 1: If $n = 1$, that would mean

$$|C(a)| = |Z(G)|.$$

But, if these groups have the same order and $Z(G) \leq C(a)$, it must be that $C(a) = Z(G)$. But, since $a \in C(a)$ and $C(a) = Z(G)$, that means $a \in Z(G)$ which contradicts our assumption.

Case 2: If $n = p$, that means

$$\frac{|G|}{|C(a)|} = 1$$

$$\Rightarrow |G| = |C(a)|.$$

Since $C(a) \leq G$, it must be that $G = C(a)$. But, this equality implies that every element in G commutes with a , making $a \in Z(G)$ which contradicts our assumption. $\square$

So, it must be that the index of $Z(G)$ cannot be prime.